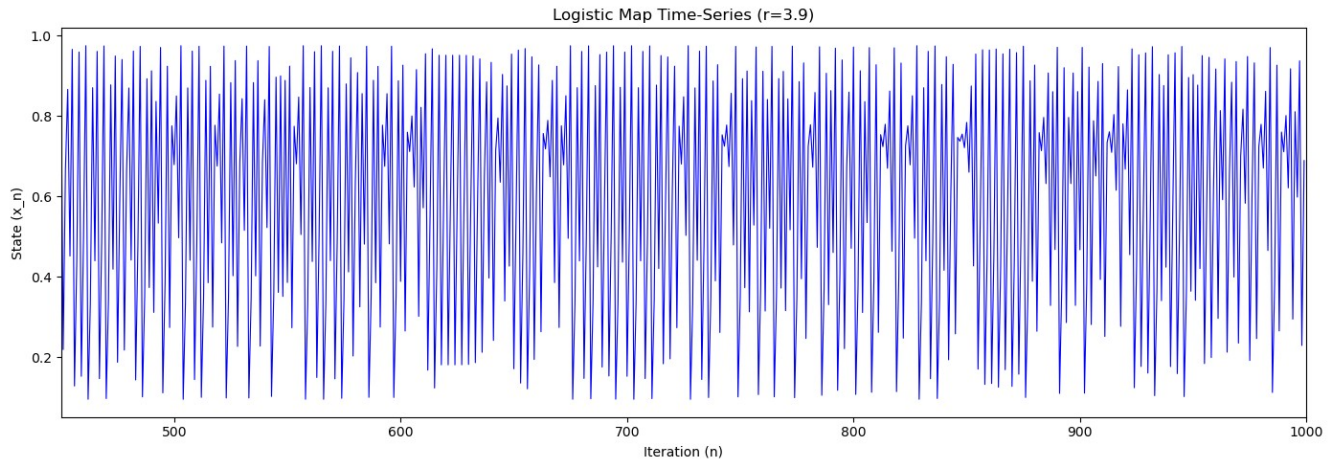


# Task (1) Report “Chaos Theory”

## Part A

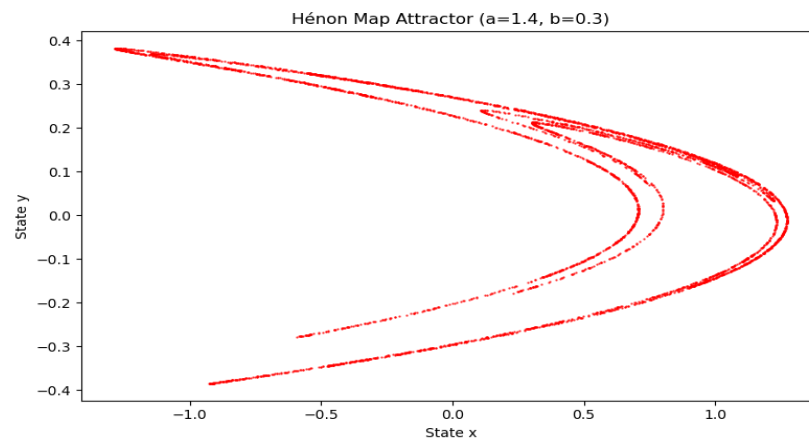
---

- A.1) Logistic Map Time Series

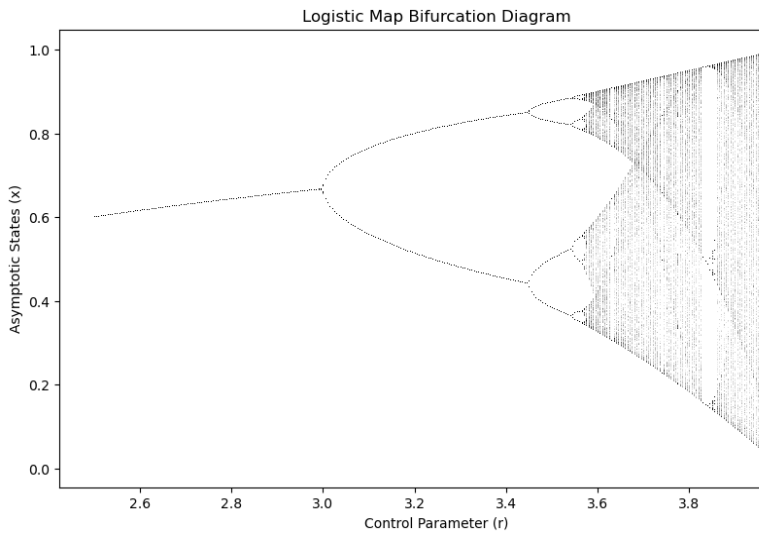


- A.2

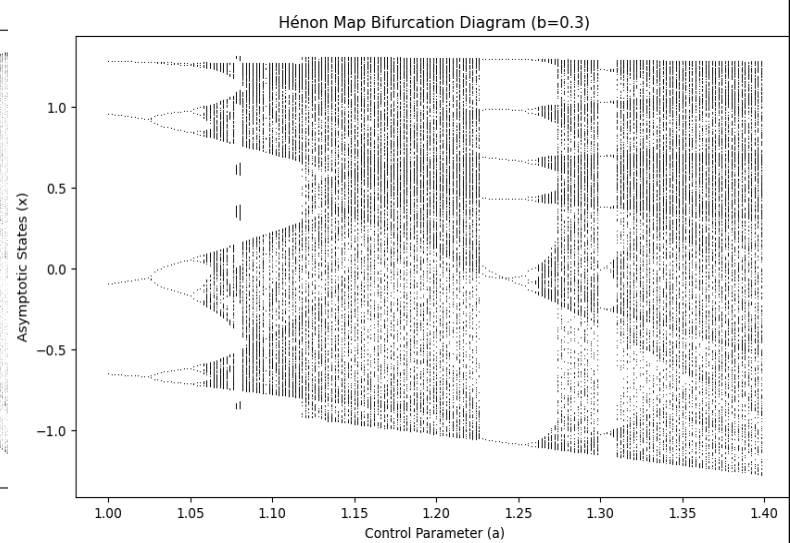
- A.2) Henon Map Attractor Plot



- A.3) Logistic Map Bifurcation Diagram



- A.4) Henon Map Bifurcation Diagram



### A.3) Logistic Bifurcation:

- It shows that the time-series is transient between (0:3) “ $r$ ” value range, while between (3 : 3.5) the time-series starts becoming periodic, but from (3.5 : 4) the time series becomes chaotic and it is becoming more and more chaotic as we increase the control parameter “ $r$ ”. (so the “ $r$ ” value has direct relation with the chaos level (complexity) of the resulting time-series). But there is a special case between (3.82 : 3.85) the system returns to be transient then periodic in this small interval.

### A.4) Henon Bifurcation:

- it is different from the logistic map, here the control parameter ( $a$ ) is not uniformly related with the chaos level, as we can see from the value range of “ $a$ ” between (1 : 1.06) the data is starting transient (stable) and turning periodic, then it becomes chaotic between (1.06 : 1.225) it becomes very chaotic. then the weird thing happens here, between (1.226 : 1.27) it returns transient and periodic again, then after that it returns chaotic, so the ( $a$ ) control parameter is not in a direct relation with the chaos level of the system.

# Part B

## B.1) Lorenz:

1. Equations:
- $\dot{x} = \sigma(y - x)$

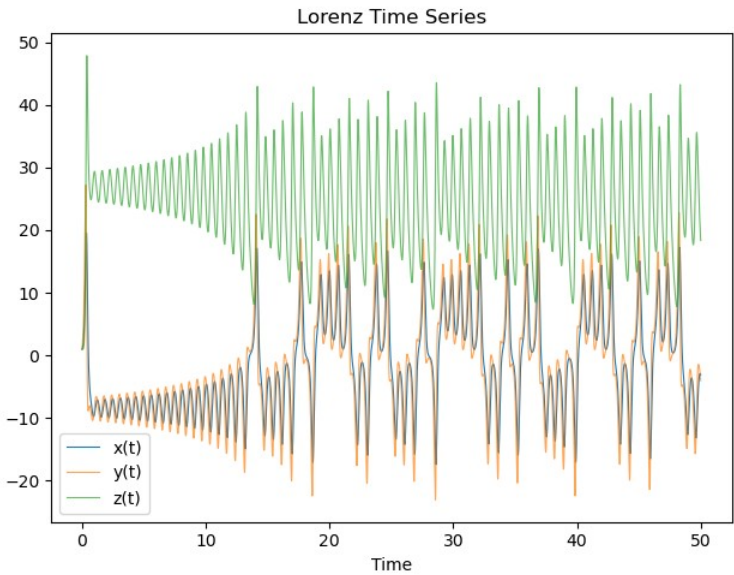
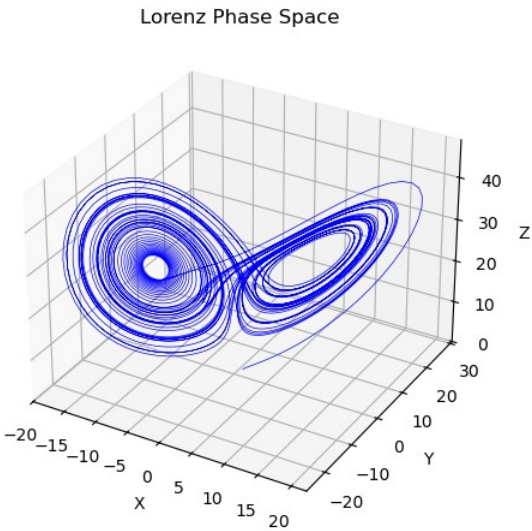
◦  $\dot{y} = x(\rho - z) - y$

◦  $\dot{z} = xy - \beta z$

2. Settings Table:

| Settings                      | Values                     |
|-------------------------------|----------------------------|
| Parameters (sigma, rho, beta) | Sigma:10, Rho:28, Beta:8/3 |
| Initial Conditions            | (x=1, y=1, z=1)            |
| Time step (dt)                | 0.01                       |
| Length                        | 50                         |
| Solver                        | RK4(fixed step)            |

3. Plots:



B.2) Rossler:

1. Equations:
- $\dot{x} = -y - z$

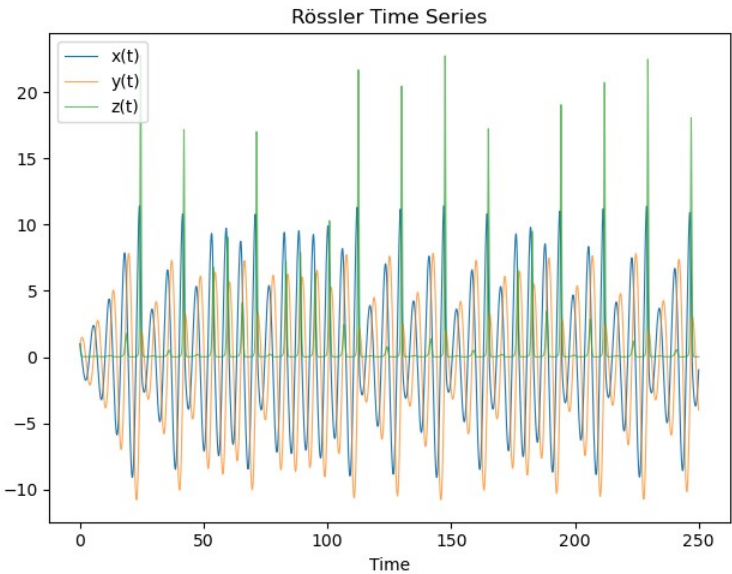
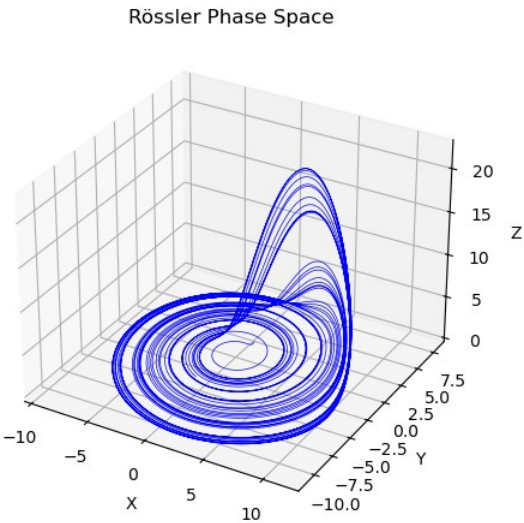
◦  $\dot{y} = x + a \cdot y$

◦  $\dot{z} = b + z(x - c)$

2. Settings Table:

| Settings             | Values                 |
|----------------------|------------------------|
| Parameters (a, b, c) | A: 0.2, b: 0.2, c: 5.7 |
| Initial Conditions   | (x=1, y=1, z=1)        |
| Time step (dt)       | 0.01                   |
| Length               | 250                    |
| Solver               | RK4(fixed step)        |

3. Plots:



B.3) Chen:

1. Equations:
- $\dot{x} = a(y - x)$

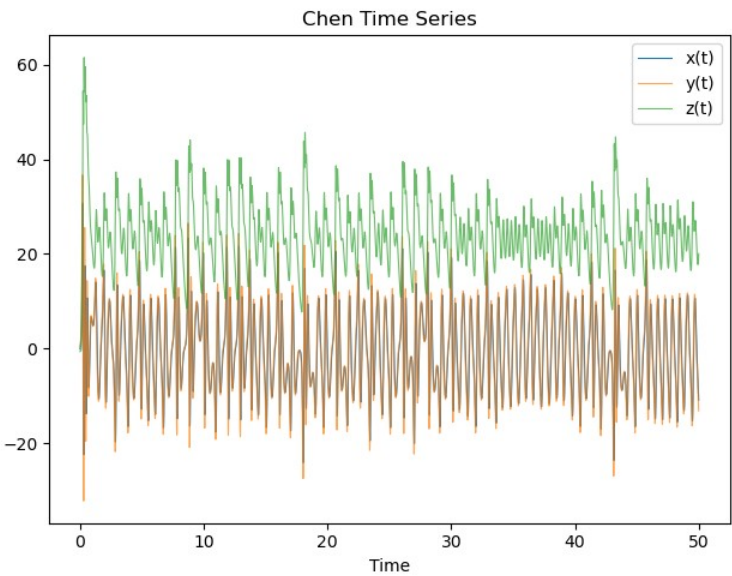
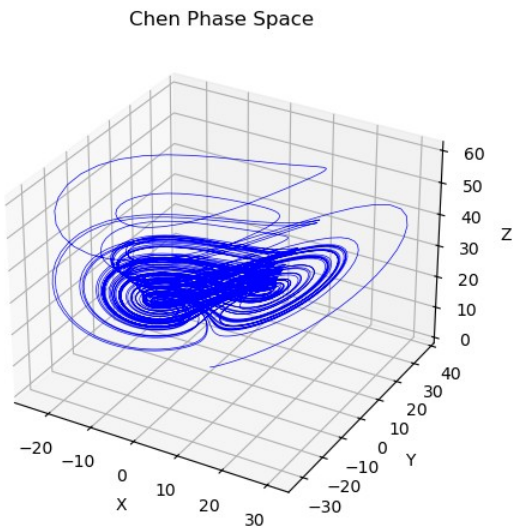
◦  $\dot{y} = (c - a)x - xz + cy$

◦  $\dot{z} = xy - bz$

2. Settings Table:

| Settings             | Values                  |
|----------------------|-------------------------|
| Parameters (a, b, c) | a: 35, b: 3, c: 28      |
| Initial Conditions   | (x=-0.1, y=0.5, z=-0.6) |
| Time step (dt)       | 0.01                    |
| Length               | 50                      |
| Solver               | RK4(fixed step)         |

3. Plots:



B.4) Hyperchaotic Rossler:

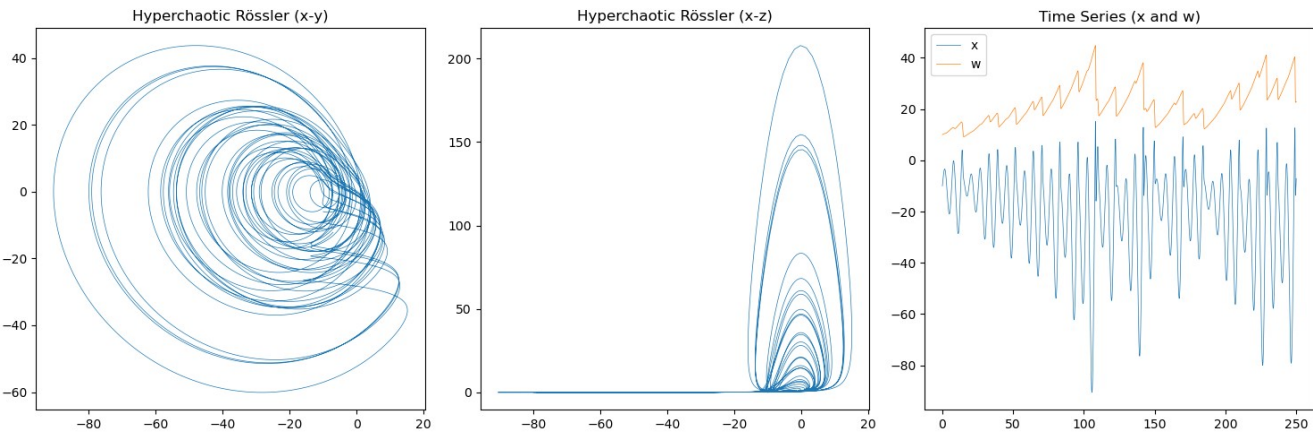
1. Equations:

◦  $\dot{x} = -y - z \quad \dot{y} = x + a \cdot y + w \quad \dot{z} = b + xz \quad \dot{w} = -c \cdot z + d \cdot w$

2. Settings Table:

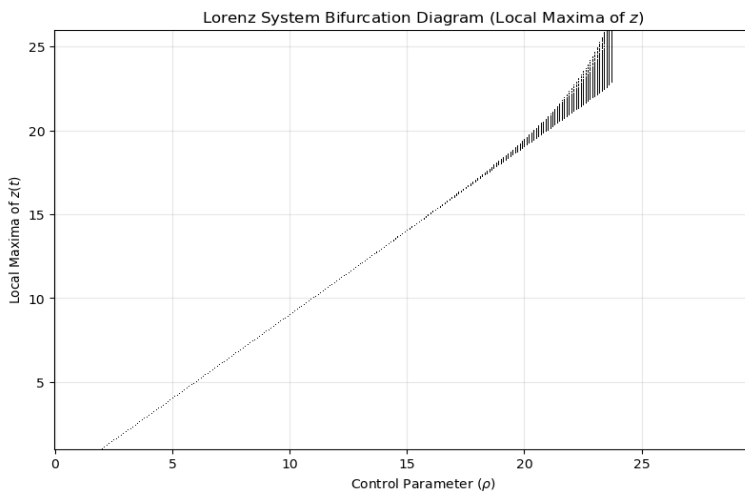
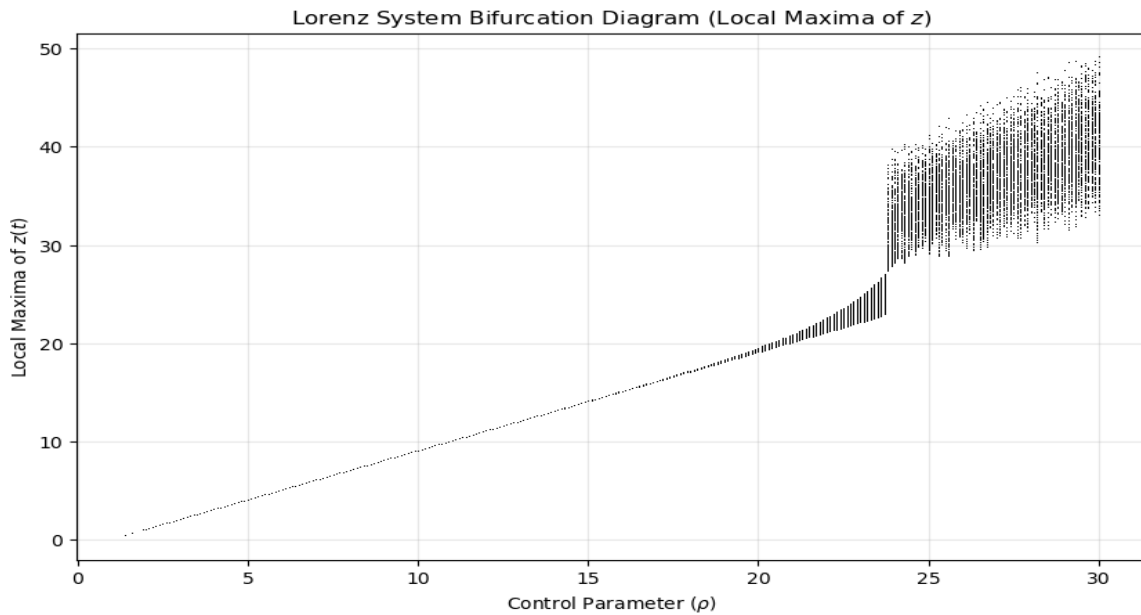
| Settings             | Values                         |
|----------------------|--------------------------------|
| Parameters (a, b, c) | A: 0.25, b: 3, c: 0.5, d: 0.05 |
| Initial Conditions   | (x=-10, y=-6, z=0, w=10)       |
| Time step (dt)       | 0.01                           |
| Length               | 250                            |
| Solver               | RK4(fixed step)                |

3. Plots:

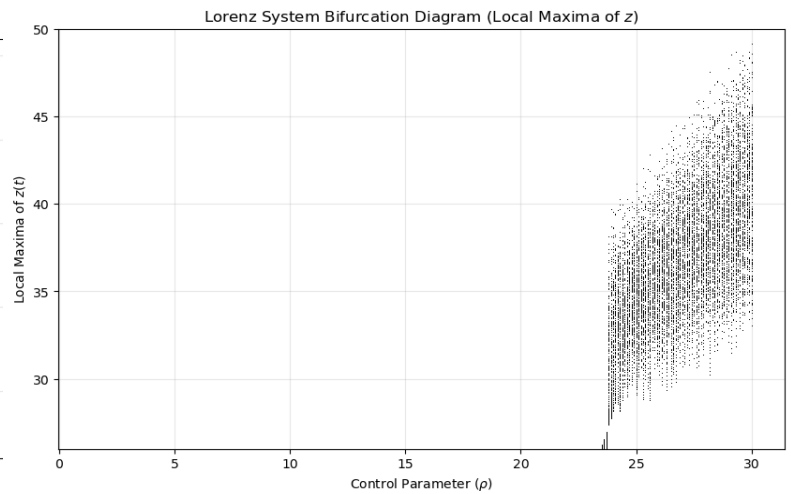


# Part C

## C.1) rho-sweep plot for Lorenz:



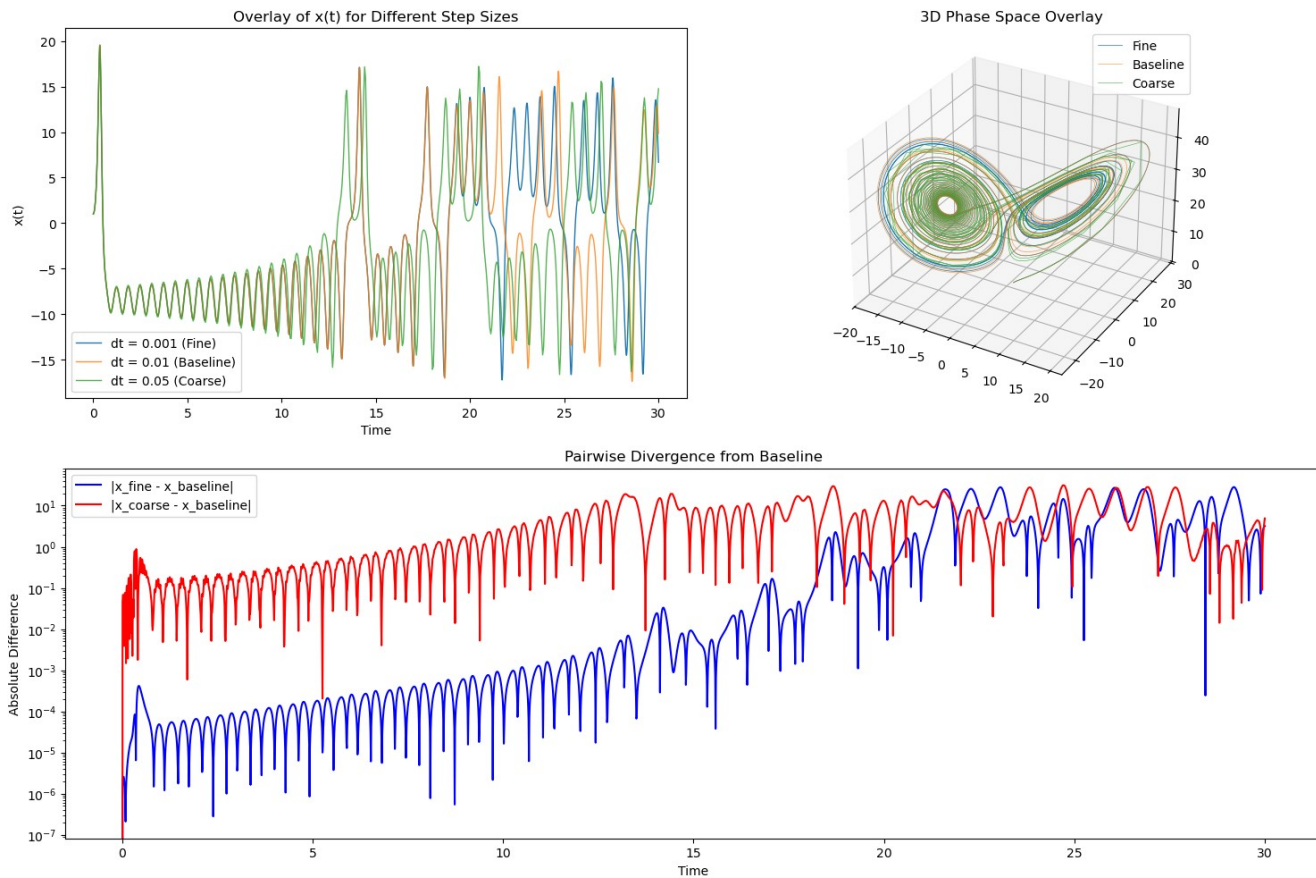
a) This section of the plot show the flow of the system from being transient into periodic



b) This section of the plot show the region where the system starts becoming chaotic.

- This plot shows the number of the unique local maxima values at each time-step in the Lorenz time-series for each value of the Rho ( $\rho$ ) parameter, left plot shows that Rho values between (0 : 24) has low number of unique values, which implies data is more periodic, while right plot with Rho value range between (24 : 30) has larger number of unique local maxima values which means data became chaotic and not following any periodic pattern.

## C.2) Step size sensitivity:



- For the  $|x_{\text{fine}} - x_{\text{baseline}}|$  plot, the error was starting with low value because the fine step size is more accurate as it is much smaller size (and its value is close to that of the baseline), but when time passes the difference in the time step size - (despite being smaller) - build up causing a very large increase in the error which is obvious in the "Pairwise Divergence" plot that between time step 10 and 20, the error increased exponentially, while the  $|x_{\text{coarse}} - x_{\text{baseline}}|$  is already starting from a high error and keep increasing because the coarse step size is larger (and large difference between the coarse and baseline steps) which being less accurate and the butterfly effect occurs strongly with it. But note that the error margin increase to a certain bound, because the systems are bounded, so the error is also bounded.
- Also in the " $x(t)$  overlay" plot, the effect of the different step sizes is so clear, at the beginning there is a tiny difference between the three systems, but the butterfly effect appears obvious. As from time step 0 to 10, all the three systems are outputting identical values and overlapping strongly, but after that, they start diverging from each other exponentially, which is a result of the tiny difference between them in the step size (considered like the initial conditions change).

## Final Conclusion:

Based on the simulation results, discrete chaos evolves in sudden, disconnected jumps from one state to the next, whereas continuous chaos flows smoothly along continuous and oscillating paths over time. Furthermore, standard chaotic systems like the Lorenz model diverge unpredictably in just one direction, creating clear, distinct boundaries like the 3D butterfly shape. In contrast, Hyperchaotic systems diverge in multiple directions at the same time, resulting in much messier and highly complex data patterns that require four dimensions to properly capture.